INCH-POUND

MIL-M-28787/79A
AMENDMENT 3
16 December 1992
SUPERSEDING
AMENDMENT 2
14 November 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LOGIC UNIT, ARITHMETIC, DIGITAL, KEY CODE RDH

This amendment forms a part of Military Specification MIL-M-28787/79A(EC), dated 31 July 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 3, 4, AND 5

1.2, Propagation delay time (t_{PLH}) and (t_{PHL}): Delete in their entirety.

PAGE 6

1.3d: Delete in its entirety and substitute:

"d. The module also includes a look-ahead carry generator (LCG) capable of generating a carry across four binary adders or groups of adders. The LCG can be cascaded to perform full look-ahead across N-bit adders. Positive logic equations for the LCG are:

(1) $C_{N+X} = G0 + POC_N$.
(2) $C_{N+Y} = G1 + P1G0 + P1P0C_N$.
(3) $C_{N+Z} = G2 + P2G1 + P2P1G0 + P2P1P0C_N$.
(4) $\overline{G} = \overline{G3 + P3G2 + P3P2G1 + P3P2P1G0}$.
(5) $\overline{P} = \overline{P3P2P1P0}$."

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PAGE 12

*TABLE IV, Test conditions D1, D2-D13, E1, E2-E20, and G1-G23, Test type column: Delete "X" and substitute "S".

PAGE 15

*TABLE IV, Test conditions Q45-Q48, Acceptance requirements 25° ± 5°C, Max column: Delete "35.5" (two places) and substitute "38.5" (two places).

*TABLE IV, Test conditions Q45-Q48, Operating temperature 0° to 60°C, Max column: Delete "38.0" (two places) and substitute "41.0" (two places).

PAGE 23

4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, D, E, and G are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, and G where applicable, followed by the remaining tests in the sequence."

PAGE 24

*4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table IV shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F and H through V shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

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*4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F and H through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table IV. Following the return of the module to 25 ± 5°C, electrical tests F and H through V shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table IV.

PAGE 32

FIGURE 7: Delete and substitute new figure 7 as printed on page 5 of this amendment.

PAGE 35

FIGURE 8, Item PO1-2 box: Delete "5.5V ± 1%" and substitute "4.0V ± 1%".

PAGE 43

TABLE X, note 2, second line: Delete "voltmeter (V03)" and substitute "item MO2".

PAGE 44

FIGURE 11: Delete and substitute new figure 11 as printed on page 6 of this amendment.

PAGE 47

FIGURE 12: Delete and substitute new figure 12 as printed on page 7 of this amendment.

PAGE 50

FIGURE 13: Delete and substitute new figure 13 as printed on page 8 of this amendment.

PAGES 72 AND 73

*TABLE XVIII, Test conditions Q45, Q46, Q47, and Q48, Max column: Delete "35.5" (four places) and substitute "38.5" (four places).

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PAGE 115

*FIGURE 25: Delete and substitute new figure 25 as printed on page 9 of this amendment.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

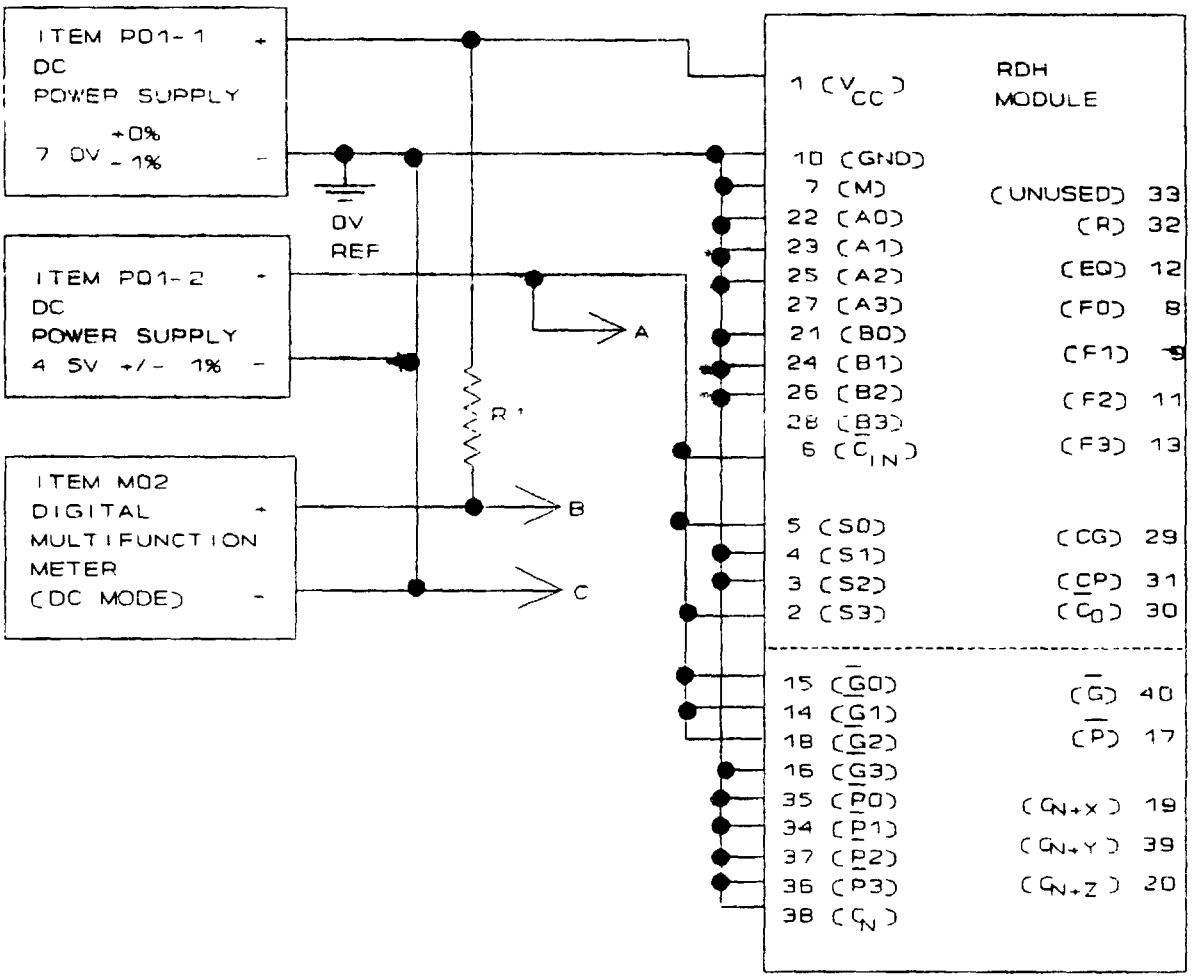
Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

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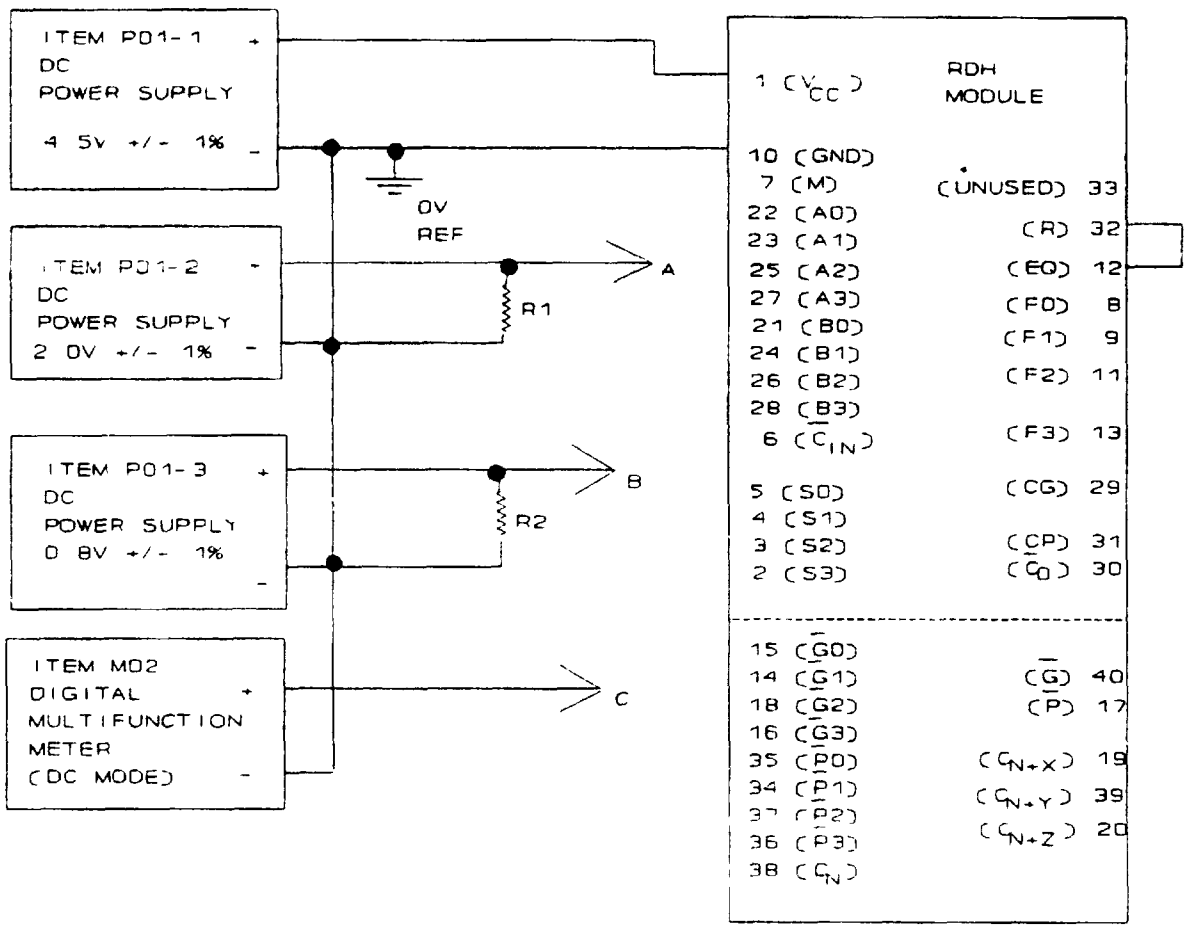
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NOTE: R1 = 249Ω ± 1%, 1/4W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table VI.

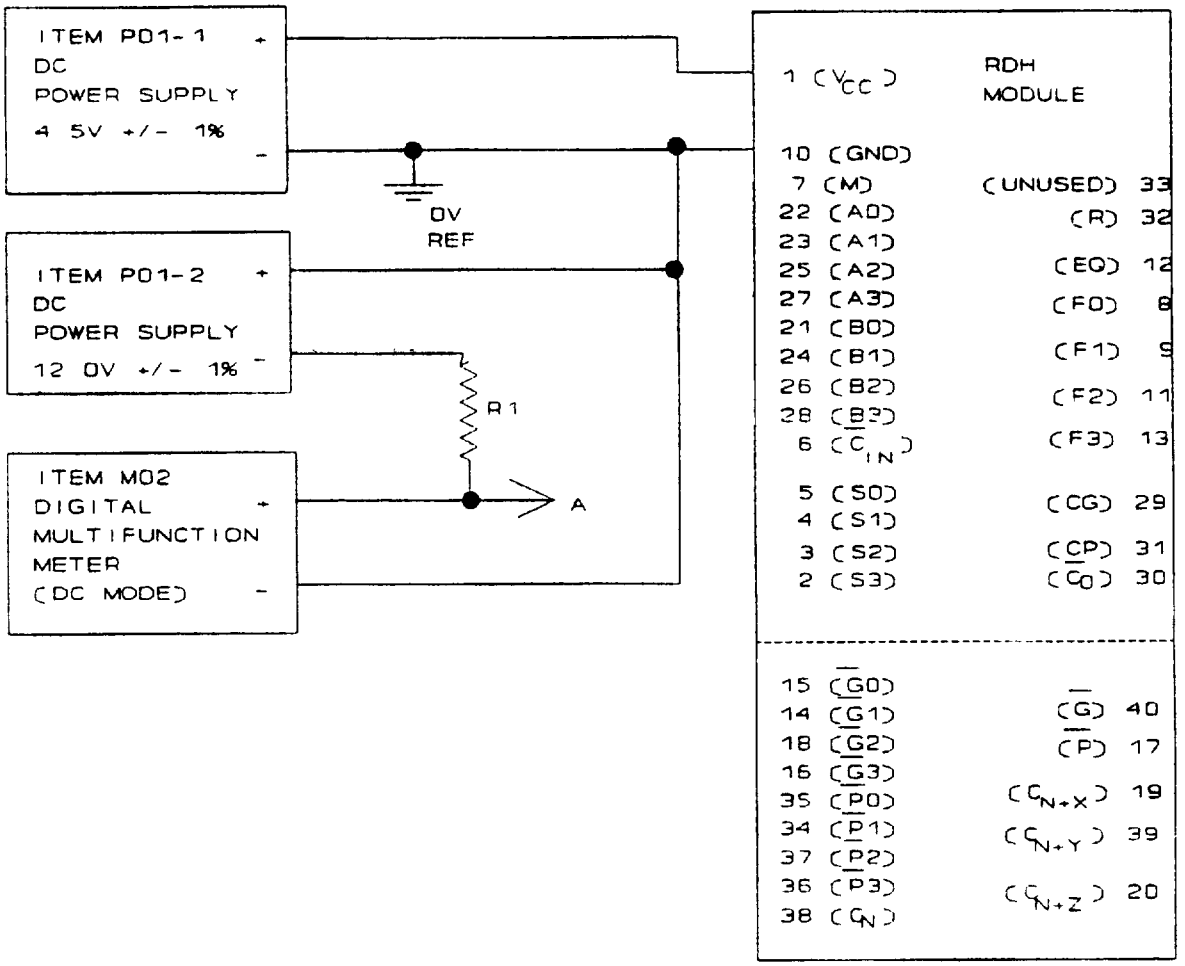
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NOTE: R1 = R2 = 51Ω ± 5%, 1/2W.

FIGURE 11. Functional test circuit for table X.

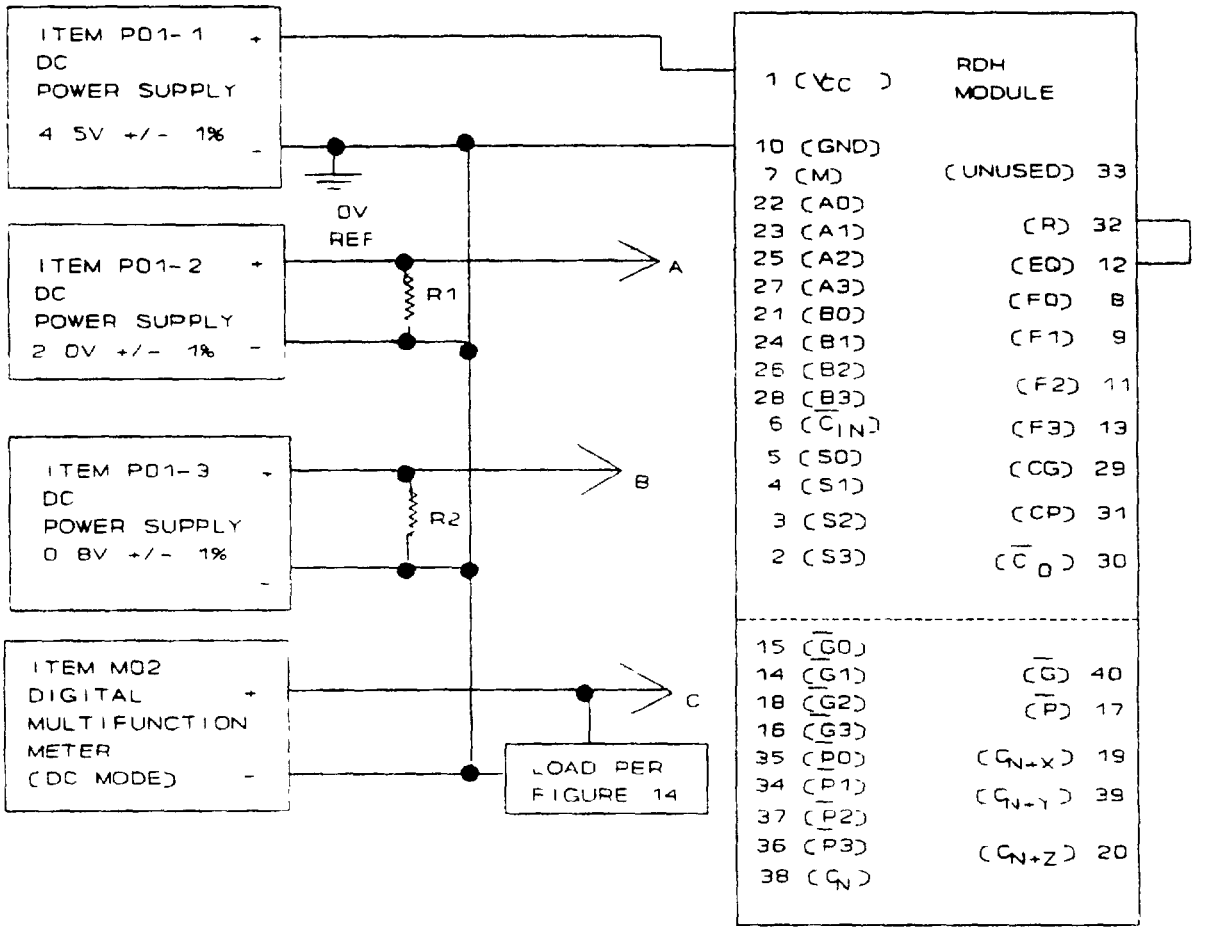
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NOTE: R1 = 590Ω ± 1%, 1/4W.

FIGURE 12. Input clamp diode voltage test circuit for table XI.

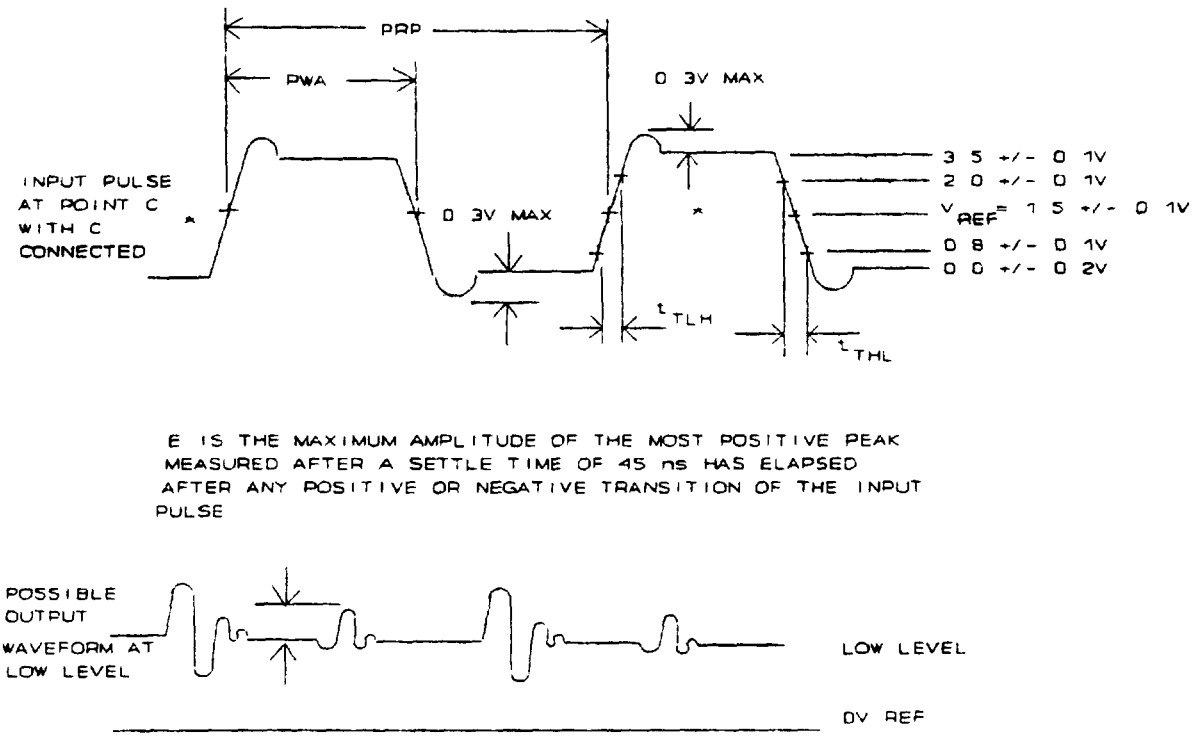
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NOTE: R1 = R2 = 51Ω ± 5%, 1/2W.

FIGURE 13. Static output voltage (low-level and high-level) test circuit for tables XII and XIII.

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NOTES:

1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 250 ± 25 ns.
   PRP = 500 ± 50 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

FIGURE 25. Output crosstalk duration test circuit waveforms for table XIX.

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